3D Stokes Polarimetry of Dipolar Near-Fields

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Abstract: Dipolar emitters constitute fundamental building blocks in nanophotonics, yet their polarization behavior outside the far-field approximation remains insufficiently characterized. In this work, we introduce the three-dimensional Stokes formalism as a unified framework to describe the complete polarimetric landscape of dipolar radiation across the near-, intermediate-, and far-field regimes.

Keywords: Stokes, polarimetry, dipolar fields.

Motivation

- Dipolar emitters have complex polarization structures, enabling many applications [1, 2].
- Existing formalisms (2D Stokes or spin density) do not capture the full picture of dipolar polarization.

Theoretical foundation

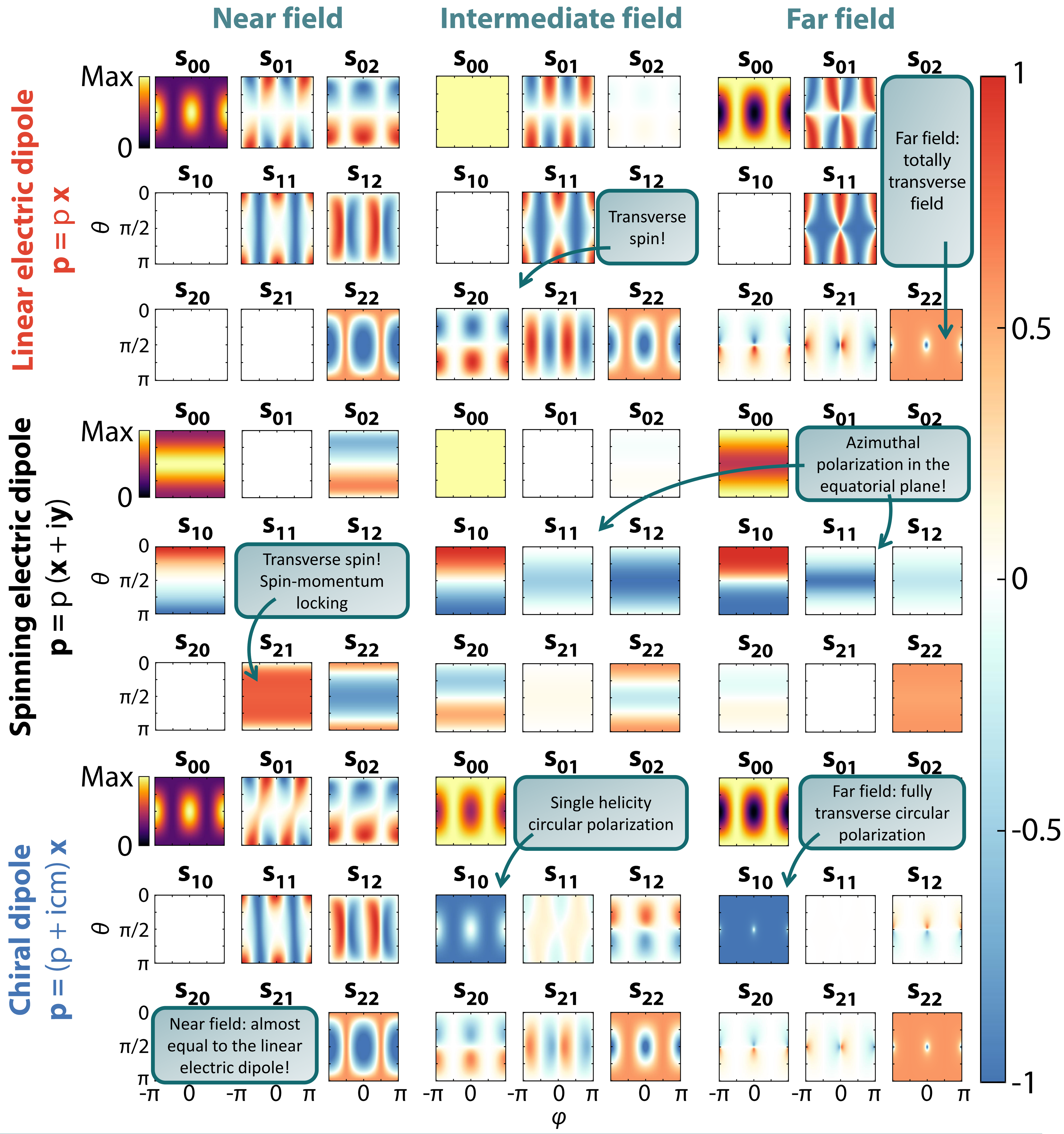
- A non-paraxial field can be polarimetrically described by a set of 3D Stokes parameters, derived from the 3D polarization matrix [3].
- To match typical experimental situations and provide meaningful interpretations, we propose a modified spherical coordinate system (θ, φ, r):

$$\overline{\overline{\mathbf{R}}} = \langle \mathbf{E} \otimes \mathbf{E}^{\dagger} \rangle = \begin{pmatrix} E_\theta E_\theta^* & E_\theta E_\varphi^* & E_\theta E_r^* \\ E_\varphi E_\theta^* & E_\varphi E_\varphi^* & E_\varphi E_r^* \\ E_r E_\theta^* & E_r E_\varphi^* & E_r E_r^* \end{pmatrix}$$

- The interpretation of these 3D Stokes parameters is:

	Intensity	Linear polarization	Linear polarization
	s_{00}: $I_\theta + I_\varphi + I_r$	s_{01}: $I_{\theta,45} - I_{\varphi,45}$	s_{02}: $I_{\theta,45} - I_{r,45}$
Circular polarization	s_{10}: $I_{\theta\varphi,R} - I_{\theta\varphi,L}$	s_{11}: $I_\theta - I_\varphi$	s_{12}: $I_{\varphi,45} - I_{r,45}$
Circular polarization	s_{20}: $I_{\theta r,R} - I_{\theta r,L}$	s_{21}: $I_{\varphi r,R} - I_{\varphi r,L}$	s_{22} (Field transversality): $(I_\theta + I_\varphi) - I_r$

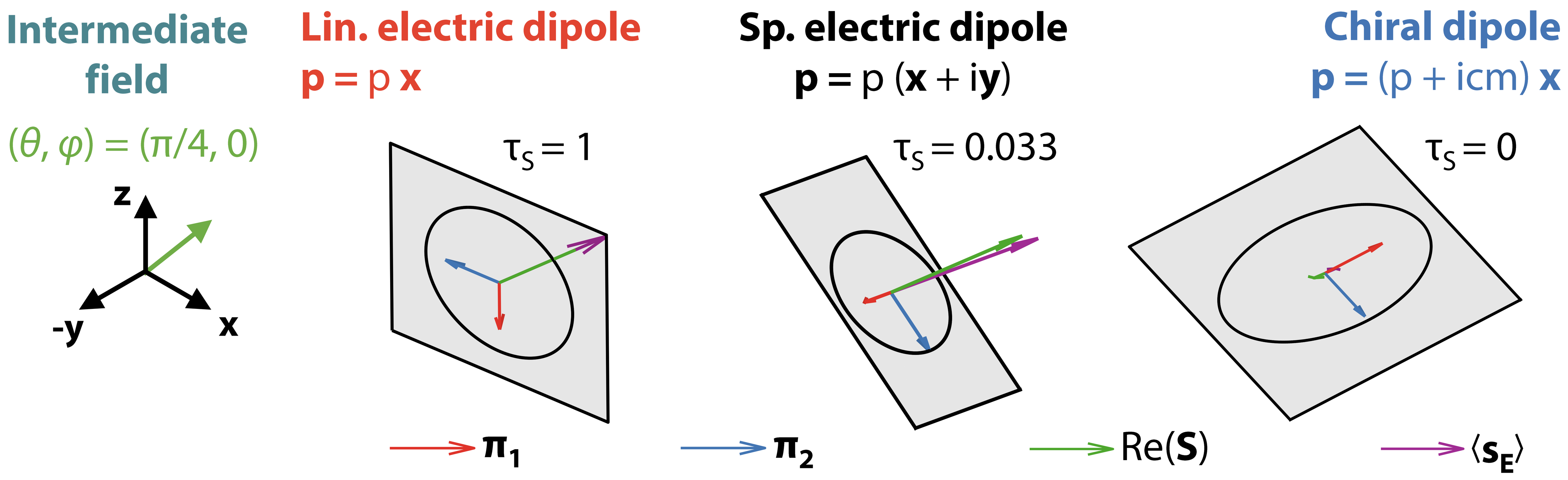
3D Stokes parameters in dipolar fields



Intrinsic reference frame and spin-transversality index

- Since dipolar fields are pure polarimetric states, they can always be described by a local 2D polarization ellipse.
- This ellipse lies in the "intrinsic reference frame".
- We can find this ellipse via a Gram-Schmidt process, and then analyze the resulting frame with respect to energy propagation (real part of the Poynting vector) and spin $\langle \mathbf{s}_\mathbf{E} \rangle$.
- We use a "spin transversality index":

$$\tau_S = \sin\theta_C = \sqrt{1 - \left[\frac{\langle \mathbf{s}_\mathbf{E}\rangle \cdot \mathrm{Re}(\mathbf{S})}{|\langle \mathbf{s}_\mathbf{E}\rangle|\,|\mathrm{Re}(\mathbf{S})|}\right]^2}$$



References

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[2] K. Y. Bliokh et al., *Nat. Phot.* **9,** 12 (2015).
[3] J. J. Gil and R. Ossikovsky, *Polarized Light and the Mueller Matrix Approach.* CRC Press (2016).

Conclusions

- 3D Stokes polarimetry provides a complete local description of dipolar polarization:
  - Recognizes not only circular vs linear polarizations, but also their directions.
  - Accurately reproduces predictions from other works [1,2].
- The intrinsic reference frame and the spin transversality index reveal fundamental differences between dipoles underlying common polarization structures:
  - Linear electric dipoles have totally transverse spins.
  - Spinning dipoles transition from near-field transverse spin to far-field longitudinal spin.
  - Chiral dipoles have totally longitudinal spins.

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